CiteAudit: You Cited It, But Did You Read It?
A Benchmark for Verifying Scientific References in the LLM Era

Web Application: http://www.checkcitation.com/

Zhengqing Yuan[1] Kaiwen Shi[1] Zheyuan Zhang[1] Lichao Sun[2] Nitesh V. Chawla[1] Yanfang Ye[1*]

[1]University of Notre Dame [2]Lehigh University

Abstract

Scientific research is the fundamental driver of human societal progress, and proper citation is vital for attribution and research integrity. However, the rise of large language models (LLMs) has introduced a new integrity risk: fabricated references that appear plausible but correspond to no real publications. Recent analyses have uncovered such hallucinated citations even in submissions and accepted papers at major machine learning venues, underscoring growing vulnerabilities in peer-review workflows and raising concerns about the credibility of scholarly discourse. At the same time, rapidly expanding reference lists render manual verification infeasible, while existing automated tools remain fragile to the noise and formatting variability of real-world citation data and lack standardized, transparent evaluation.

This paper addresses these challenges by introducing the first comprehensive benchmark and detection framework for hallucinated citations in scientific writing. We design a multi-agent verification pipeline that decomposes citation checking into coordinated stages—claim extraction, evidence retrieval, passage matching, contextual reasoning, and calibrated judgment—enabling robust assessment of whether a cited source genuinely supports its associated claim. We further construct a large-scale, human-validated dataset spanning diverse domains and citation types, together with unified evaluation protocols for citation faithfulness and evidence alignment. Experiments with state-of-the-art LLMs reveal substantial citation-related errors and demonstrate that our multi-agent framework significantly outperforms existing baselines in both accuracy and interpretability. Our work provides the first systematic infrastructure for auditing citations at scale in the LLM era, offering practical tools for researchers, reviewers, and publishers to strengthen the trustworthiness of scientific references.

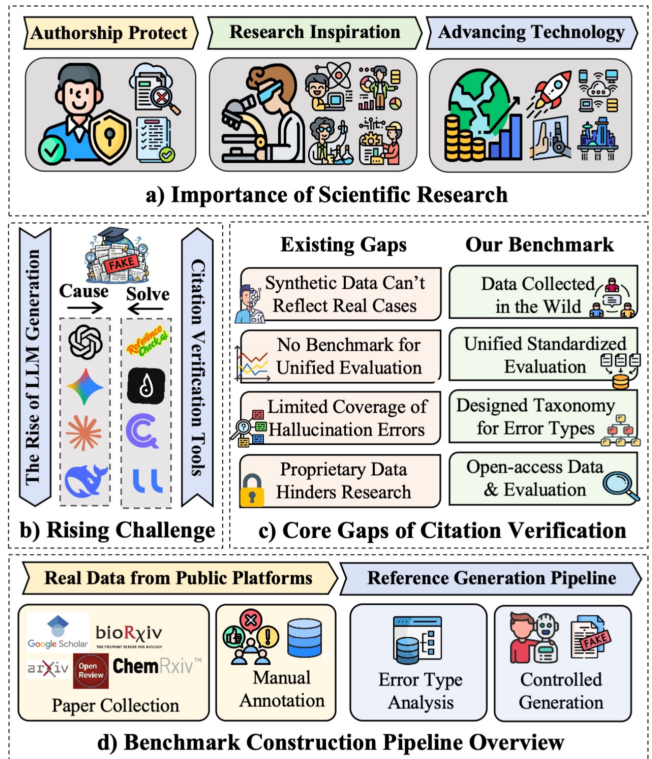

Figure 1: Motivation overview of our citation hallucination benchmark: existing gaps in closed-source citation checking tools and the unified, reliable evaluation framework enabled by our benchmark.

1 Introduction

Scientific research constitutes the most critical engine of human progress, and the protection of authorship represents a fundamental respect for scholarly contributions as well as a vital source of intellectual inspiration. When citations are missing, inaccurate, or fabricated, the resulting break in the evidence chain can obscure logical dependencies, weaken argumentation, and jeopardize academic integrity at the level of both individual papers and the broader literature [10, 22, 26]. However, the rapid adoption of large language models (LLMs) [28] and other generative systems [29] has introduced a qualitatively new risk: the automatic creation of bibliographic entries that have no counterpart in the scholarly record. In contrast to conventional citation mistakes, such as incomplete metadata or minor typographical errors, hallucinated citations are entirely fabricated references that nevertheless resemble legitimate academic works. Recent investigations have documented such cases across prominent machine learning conferences, revealing hallucinated references in multiple submissions [6]. Independent reports further indicate that similar problems have surfaced even in accepted papers at leading forums such as NeurIPS and ACL [7, 21]. These incidents compromise multiple layers of the research process: they impede reviewers' ability to assess evidence, expose co-authors to inadvertent integrity violations, and weaken the reliability of the publication ecosystem as a whole, with downstream implications for reproducibility and the credibility of scientific discourse.

The growing scale of scholarly publishing further complicates this problem: reference lists have expanded rapidly across disciplines [12], making thorough manual verification unrealistic for

*Corresponding author: yye7@nd.edu

reviewers, editors, and co-authors. This has motivated the development of automated citation-auditing tools [1, 3, 5]. However, **prior works generally demonstrate two major gaps**: 1) Citation verification inevitably relies on retrieving information from external sources, yet the inherent noise and formatting variability of real-world references exposes a core weakness of existing systems, which frequently misfire when citations deviate from clean, canonical forms. 2) Most of these systems are proprietary, where they neither released their mechanism nor, more importantly, a large-scale, standardized and reproducible benchmark for hallucinated citation detection.

To bridge those two gaps, in this paper, we introduce, to the best of our knowledge, the first comprehensive benchmark and detection framework for hallucinated citations in scientific manuscripts. Our contributions lay the groundwork for scalable tools that can support reviewers, editors, and automated review systems in upholding scholarly rigor in the LLM era. Specifically, we present a multi-agent framework and accompanying benchmark for verifying citation faithfulness in scientific text. **To cope with the first challenge**, our system decomposes citation verification into cooperative roles: a Claim Extractor identifies citation-bearing statements, a Retriever gathers candidate evidence from cited sources, an Evidence Matcher aligns claims with passages, a Reasoner adjudicates support/contradiction, and a Judge produces calibrated verdicts with explanations. This pipeline enables fine-grained assessment of whether a cited reference truly supports the claim it is attached to. **To address the second challenge,** we extend this framework to construct a large-scale dataset spanning diverse domains and citation types (supporting, contrasting, methodological, and background references), with human-validated labels and model-generated explanations for transparency. We introduce evaluation protocols and metrics that measure citation faithfulness, evidence alignment, and verdict consistency across models. Experiments over leading LLMs reveal substantial rates of citation errors, including unsupported claims, misattributed findings, and subtle semantic drift between claims and sources. Our analysis shows that multi-agent verification significantly improves detection accuracy and interpretability compared to single-model baselines. This work provides the first systematic infrastructure to audit citations at scale in the LLM era, offering a practical tool for researchers, reviewers, and publishers to assess and improve the trustworthiness of scientific references. Our Contributions can be summarized as follows:

- **Benchmark.** We release the first large-scale, standardized benchmark for hallucinated citation detection, covering diverse domains and citation types with human-validated labels and unified evaluation protocols.
- **Framework.** We introduce a multi-agent verification pipeline that separates claim extraction, retrieval, matching, reasoning, and judgment, enabling robust citation checking under noisy and heterogeneous real-world formats.
- **Findings.** Through extensive experiments on state-of-the-art LLMs, we uncover pervasive citation errors and show that our framework yields stronger accuracy and interpretability than existing baselines.

2 Related Works

2.1 Reference Hallucination Detection

Reliably identifying AI-generated hallucinated content has become an increasing concern for both academia and industry [8, 15]. Among them, one of the most concerning risks lies in hallucinations in academic writing, where LLMs generate non-existent references [10]. Such errors undermine scholarly trust and threaten the integrity of scientific communication [8, 25]. To verify the authenticity of academic references, some works [1, 3, 5] have emerged to audit references by parsing citation strings and matching them against external bibliographic databases. Yet retrieval-based citation checking pipelines remain brittle to noise and variability inherent in real-world references, thus limiting their performance. To address this, more recent systems [2, 4] adopt fuzzy matching strategies that compare citation fields against retrieved records using token-level similarity rather than exact string matching, enabling detection of mutated or incomplete references. However, these approaches still fundamentally reduce verification to field-level similarity matching, thus often failing under subtle or incomplete reference perturbations. More recent research begins to combine LLM-based reasoning models with retrieval for citation verification [16], but these early efforts rely on overly limited and homogeneous external database sources, which can lead to false positive errors in practice. Moreover, their applicability can be further challenged in realistic settings where references must be extracted and verified from complex multimodal scholarly documents.

2.2 Web Search Agent and Fact Checking

LLM-based agents have recently demonstrated strong performance on complex, long-horizon tasks by moving beyond pure text generation toward actionable decision-making pipelines that interleave reasoning with interactions in external environments [13, 14, 23, 30, 32]. A key advantage of agentic systems over standalone foundation models is tool use—the ability to invoke external modules to acquire up-to-date evidence, execute operations, and reduce reliance on parametric memory. A representative and widely adopted form of tool use is web search [17, 19, 27], which enables agents to ground answers in retrieved evidence and thereby mitigate hallucinations [31, 33]. Early works have further applied web search agents to fact-checking settings, demonstrating their effectiveness in evidence-based misinformation detection [24]. In the context of citation verification, these advances in web search agent architectures motivate moving beyond approaches that rely solely on limited bibliographic APIs: by harnessing broader web search agents, systems can access a more comprehensive and diverse set of sources, thereby mitigating the coverage limitations inherent in API-based citation checks and improving the robustness of citation hallucination detection.

3 Benchmark

While a growing set of citation-verification systems has been developed to detect hallucinated citations in academic writing, many of these systems remain closed-source and proprietary, rendering their verification mechanisms opaque and their empirical performance irreproducible [20]. This lack of transparency hinders systematic

Table 1: Statistical overview of our citation hallucination benchmark, including both the generated benchmark and the real-world test set.

Subset	Real	Fake	Data Source
Generated Test Set	3,586	2,500	GPT, Gemini, Claude Sonnet, Qwen, Llama, etc
Real-World Test Set	2,889	467	Google Scholar, OpenReview, ArXiv, BioRxiv, etc
Overall	6,475	2,967	

advancement: without an open, standardized benchmark, it's infeasible to fairly compare methods or establish consistent evaluation protocols, thereby constraining the field's scientific advancement, highlighting the need for a controlled, comprehensive, and reproducible benchmark for citation verification.

To address this gap, we introduced CiteAudit, a benchmark grounded in hallucinated citations reported on OpenReview. Specifically, we manually screen a large collection of papers, analyze the error patterns of citation hallucinations that naturally occur in real-world scholarly writing, and accordingly propose a structured taxonomy of fake citation types, with the detailed description in Appendix A. Building on this foundation, our benchmark integrates both ecologically observed citation errors from the academic literature and controlled hallucinated references generated through principled perturbations. Benchmark statistics are shown in Table 1.

3.1 Real World Data Collection

We begin benchmark construction by collecting real-world citation entries from authentic scholarly manuscripts. Specifically, inspired by recent hallucination incidents reported at ICLR and NeurIPS, we collect citation instances from academic papers indexed in OpenReview and Google Scholar. From these sources, we systematically sample a large set of papers to obtain representative citation entries, and we systematically cross-check their title, author list, venue, year, and other bibliographic metadata against authoritative scholarly records, and label each entry as verifiably correct only when all key fields consistently match; otherwise, we mark the reference as containing genuine errors or hallucinated components at the field level.

This real-world citation benchmark provides a high-quality gold reference set that reflects naturally occurring citation mistakes, including incorrect author attributions, venue mismatches, and nonexistent references. However, this process also highlights a key limitation: manual citation verification is highly labor-intensive and difficult to scale the dataset to a sufficiently large size.

3.2 Human-Synthesized Data Generation

To overcome this scalability gap, we introduce the second component of our benchmark construction: a systematic framework for generating large-scale, controlled hallucinated citations.

Real Citation Collection. We first extract official BibTeX entries from publicly available open-access bibliographic repositories, spanning a broad spectrum of research areas and publication venues. These verified references serve as ground-truth citation records. Next, we generate hallucinated citations through targeted edits, guided by a principled taxonomy of citation hallucination types as shown in Figure 2.

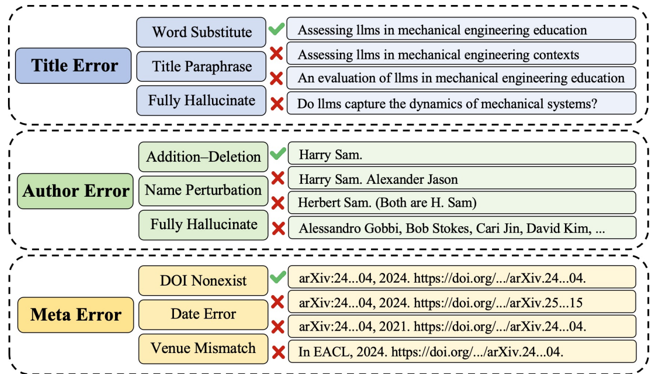

Figure 2: Taxonomy of citation hallucination types.

Title Errors Generation. Title hallucinations correspond to cases where the cited paper's title is incorrect or fabricated, while the remaining bibliographic fields remain unchanged. This error type is especially challenging because citation checkers often rely on fuzzy title matching, and minor title variations may appear plausible to both humans and automated systems. In our benchmark, we generate title errors through three complementary strategies. First, we perform keyword substitution, where core technical terms in the original title are replaced with semantically related alternatives, producing subtle but invalid variations. Second, we apply paraphrasing-based perturbations, where the title is rewritten into a fluent alternative expression that preserves topical coherence but does not correspond to any existing publication. Third, we introduce topic-conditioned fabrication, where GPT-4o-mini synthesizes a realistic-looking title within the same research area, ensuring that the hallucinated reference remains highly plausible despite being nonexistent. Together, these three strategies form an increasing spectrum of title hallucination severity, capturing hallucinations where LLMs generate citations that appear semantically appropriate but fail existence verification.

Author Error Generation. Author hallucinations occur when the author list of a citation is partially or fully incorrect. Unlike superficial formatting errors, such mistakes directly distort scholarly attribution, potentially crediting nonexistent researchers or misassigning contributions, thereby undermining academic integrity and the reliability of citation-based verification. We construct author errors via four perturbation operations. First, we simulate authorship perturbations through redundant author additions that insert nonexistent names into correct author lists and through author deletions that remove valid authors, thereby producing incomplete attribution. Then, we apply name-level perturbations to the existing author list, introducing realistic identity inconsistencies through edits to author name strings (e.g., swapping given and

Table 2: Chi-square comparison of GPTZero fake-citation detection between generated and real-world datasets.

Dataset	Pred Fake	Pred Real	Total
Generated Test Set Fake Citations	1809	691	2500
Real-World Test Set Fake Citations	57	22	79
$\chi^2 = 0.002,\ p = 0.97\ (df = 1)$			

family names) while preserving the overall author set structure. Finally, we construct fully synthetic author lists, where the entire set of authors is fabricated rather than derived from the original reference. These author perturbations enable systematic evaluation of citation checkers' sensitivity to identity-level inconsistencies.

Metadata Error Generation. Metadata hallucinations involve incorrect bibliographic fields beyond title and authors, such as venue names, publication years, or persistent identifiers. These errors reflect cases where an LLM recalls a paper approximately but misattributes its publication context. We generate metadata errors by perturbing key BibTeX fields. Venue errors are constructed by replacing the true conference or journal name with a related but incorrect outlet. Year errors are introduced by shifting publication dates to plausible but invalid alternatives. In addition, we generate DOI and identifier hallucinations, where DOI strings are fabricated or mismatched, simulating errors that cannot be detected through surface-level text similarity alone. Metadata errors therefore capture hallucination patterns where citations appear structurally complete but fail bibliographic consistency checks.

Our generated benchmark exhibits error characteristics that are highly consistent with those observed in real-world citation hallucinations. As shown in Table 2, we compare the error distribution of GPTZero on hallucinated citations between the generated benchmark and the real-world set. A chi-square test shows no significant difference ($\chi^2 \approx 0.002$, $p \approx 0.97$), indicating nearly identical behavior across the two settings and supporting the fidelity of our generation framework in simulating real-world citation hallucination patterns. Additional statistics and breakdowns of the generated datasets are provided in Appendix B.

3.3 Data Annotation

To ensure benchmark reliability, both the real-world citation entries collected in the data collection stage and the hallucinated citations generated through perturbation are subjected to a rigorous multi-step annotation and verification pipeline. We adopt a screening process that combines automated evidence retrieval with careful human validation. First, a web-search-based model is employed to retrieve corresponding online evidence for each citation, linking the bibliographic fields to relevant publication pages or authoritative scholarly records. This retrieval step provides scalable support for large volumes of citations by surfacing candidate sources for verification. Subsequently, the author team manually inspects the retrieved evidence and conducts detailed cross-checking to confirm citation authenticity and resolve potential mismatches. Through this human-in-the-loop verification, we ensure that each benchmark entry is assigned an accurate and high-confidence label, indicating whether it corresponds to a real reference or a hallucinated/erroneous citation. Through this combination of web-grounded retrieval and rigorous author-led validation, our benchmark offers a systematic and reliable resource for evaluating citation verification systems under both naturally occurring citation errors and diagnostically controlled hallucination scenarios. Detailed human annotation guidelines and verification procedures are provided in Appendix C.

4 Methodology

In this section, we formalize the detection of hallucinated citations as a multi-stage evidence verification problem. We introduce a decentralized multi-agent framework coordinated via a hierarchical Standardized Operating Procedure (SOP), designed to systematically audit scholarly references for both existence and metadata integrity.

4.1 Problem Formulation

Let $\mathcal{D}$ be a scientific document containing a set of citation strings $\mathcal{R} = \{r_1, r_2, \ldots, r_n\}$. For each citation r_i, our objective is to determine a binary verdict $v_i \in \{Fake, Real\}$. We represent each citation as a structured metadata tuple $M_i = \{m_T, m_A, m_U, m_V\}$, where fields denote title, authors, URL, and publication venue, respectively. We define the verification function $\mathcal{F} : M_i \to \{0, 1\}$ based on a **Strict Consistency Criterion** S_c:

$$S_c = \prod_{k \in \{T,A,U,V\}} \mathbb{I}(m_k = \hat{m}_k) \tag{1}$$

where $\hat{m}_k$ represents the ground-truth metadata retrieved from authoritative databases, and $\mathbb{I}(\cdot)$ is the indicator function that outputs 1 if and only if there is an **exact character-level match** between the extracted field m_k and the ground truth $\hat{m}_k$, and 0 otherwise. A citation is classified as $Fake$ if no corresponding entry exists in the global scholarly graph $\mathcal{G}_{scholar}$ or if $S_c = 0$.

4.2 Collaborative Multi-Agent Pipeline

Our framework instantiates five specialized agents, each governed by a restricted action space and a specific role in the verification pipeline.

Extractor Agent ($\mathcal{A}_{ext}$): The verification pipeline is initiated by $\mathcal{A}_{ext}$, which acts as a vision-integrated structural parser. Instead of traditional linguistic analysis, this agent orchestrates high-precision OCR tools (e.g., Nougat or PyMuPDF) to ingest raw text and visual coordinates from the PDF manuscript. The LLM then performs a schema-constrained transformation to map these unformatted strings into an immutable metadata set $M_i = \{m_T, m_A, m_U, m_V\}$. This ensures that the original citation as presented by the author is captured with zero semantic distortion, providing the raw substrate for downstream exact-match auditing.

Dual-End Memory Agent ($\mathcal{A}_{mem}$): To minimize redundant computation, $\mathcal{A}_{mem}$ executes a semantic lookup across a dual-end knowledge base $\mathcal{K}$. We formalize the retrieval as a vector similarity function. Let $Enc(\cdot)$ be the embedding model, the agent computes the confidence score s_{mem}:

$$s_{mem}(M_i) = \max_{k \in \mathcal{K}} \left(\frac{Enc(M_i) \cdot Enc(k)}{||Enc(M_i)|| \cdot ||Enc(k)||} \right) \tag{2}$$

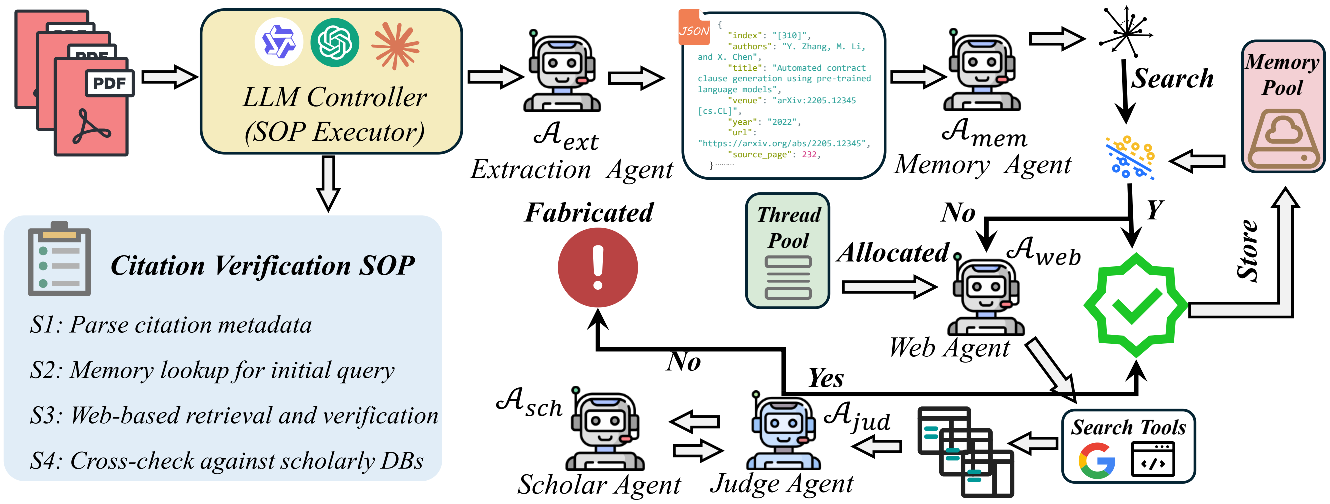

Figure 3: Overview of CiteAudit, a SOP-driven multi-agent citation verification framework.

If $s_{mem} > \tau$ (where $\tau = 0.92$), the citation is immediately verified via the "fast-path", bypassing external retrieval.

Web Search Agent ($\mathcal{A}_{web}$): In scenarios where internal memory lookup results in a cache miss, $\mathcal{A}_{web}$ is triggered to conduct exhaustive external validation. This agent interfaces with the Google Search API to identify relevant online footprints of the cited reference. Critically, rather than relying solely on search snippets, $\mathcal{A}_{web}$ executes a deep-crawling protocol to scrape the full content of the Top-5 resulting URLs. By ingesting the actual textual data from author homepages, institutional repositories, and preprint platforms, the agent ensures that the subsequent judgment is grounded in real-world evidence rather than superficial link descriptions.

Judge Agent ($\mathcal{A}_{jud}$): As the central decision engine, $\mathcal{A}_{jud}$ evaluates the alignment between the extracted metadata M_i and the retrieved evidence set $\mathcal{E}$. We define the strict verification function $\mathcal{F}_{judge}$ as:

$$\mathcal{F}_{judge}(M_i, \mathcal{E}) = \prod_{f \in \{T,A,U,V\}} \mathbb{I}(\text{ExactMatch}(M_i^f, \mathcal{E})) \tag{3}$$

where $\mathbb{I}(\cdot)$ is the indicator function. The agent acts as a gatekeeper: it returns *Real* only when all fields strictly align with the evidence from $\mathcal{A}_{web}$ or $\mathcal{A}_{sch}$.

Scholar Agent ($\mathcal{A}_{sch}$): Acting as the *Veracity Benchmark*, this agent is invoked for high-stakes validation when preliminary web evidence is insufficient. It executes targeted, low-frequency crawling of authoritative repositories (e.g., Google Scholar) to retrieve the canonical record $\hat{M}_i$.

4.3 Collaborative Multi-Agent Pipeline and Planning Model

In this section, we describe the system's execution kernel, which is orchestrated by an LLM Controller acting as the central SOP Executor. The coordination logic follows a rigorous task-allocation model governed by a Planning Model, ensuring high throughput via the Multi-Thread.

Planning Model: The orchestrator is the structural backbone of the framework. Given a PDF manuscript, the Planning Model decomposes the verification job into a sequential and parallelizable graph based on the predefined Citation Verification SOP (S1-S4). It manages the state transitions of each citation task, ensuring that resources are optimally allocated from the Thread Pool.

Formalization of SOP Execution. The coordination logic described in Stages 1-4 can be formalized as a hierarchical cascade function $\Phi(r_i)$. The Planning Model routes the verification task sequentially to optimize the trade-off between cost and accuracy:

$$\Phi(r_i) = \begin{cases} \text{Verified} & \text{if } \mathcal{A}_{mem}(r_i) > \tau \quad \text{(Stage 2)} \\ \text{Verified} & \text{if } \mathcal{A}_{jud}(r_i, \mathcal{A}_{web}) = 1 \quad \text{(Stage 3)} \\ \mathcal{A}_{jud}(r_i, \mathcal{A}_{sch}) & \text{otherwise} \quad \text{(Stage 4)} \end{cases} \tag{4}$$

This formulation ensures that the computationally expensive *Scholar Agent* ($\mathcal{A}_{sch}$) is only invoked as a fallback for the most resilient citations, adhering to the principle of minimal resource consumption.

Stage 1: Citation Metadata Extraction ($\mathcal{A}_{ext}$): The process is initiated by $\mathcal{A}_{ext}$, which maps the visual and textual data from the PDF into a structured JSON schema. This transformation converts implicit citations into explicit, verifiable metadata m_T, m_A, m_U, m_V, providing the raw data substrate for the entire pipeline.

Stage 2: Verified Memory Querying ($\mathcal{A}_{mem}$): To optimize latency, the controller routes the metadata to $\mathcal{A}_{mem}$. This agent performs a high-speed lookup in the Memory Pool. If the reference has been previously audited and marked as *Verified* (Y), the task concludes immediately. If not found (N), the planning model allocates the task to the next tier of the thread pool.

Stage 3: Web-based Content Retrieval & Consistency Audit: Uncached citations are processed by the Web Search Agent and the Judge Agent. The orchestrator allocates parallel threads to scrape high-relevance evidence from the live web. The Judge then performs the Strict Consistency Criterion check. A successful match

Table 3: Results on the generated test set. Runtime efficiency, API pricing, and verification performance of citation verification models. Prices are reported per one million tokens based on official or primary provider APIs when available. As GPTZero pricing is defined per word in subscription tiers, a coarse token-level estimate is used for comparison.

Model	Time	Price ($/1M tok)		Confusion Matrix				Metrics			
	/10 refs	In	Out	TP	FN	FP	TN	Acc	Prec	Rec	F1
Mixtral-8x7B-Instruct	2.3	0.60	0.60	1675	825	940	2646	0.710	0.641	0.670	0.655
Llama-3.3-70B-Instruct	4.9	0.88	0.88	1088	1412	381	3205	0.705	0.741	0.435	0.548
Qwen3-Next-80B-A3B	3.5	0.15	1.50	1265	1235	1370	2216	0.572	0.480	0.506	0.492
Gemini-3-Pro	36.6	0.50	3.00	1879	621	511	3075	0.814	0.786	0.752	0.769
GPT-5.2	47.1	1.75	14.00	2284	216	**0**	**3586**	0.965	1.000	0.914	0.955
GPTZero	26.3	70.00	0.00	1809	691	623	2877	0.770	0.744	0.724	0.734
Claude-Sonnet-4.5	11.3	3.00	15.00	2475	25	3364	222	0.443	0.424	0.990	0.594
Our Model	**2.3**	**0.00**	**0.00**	**2500**	**0**	167	3419	**0.973**	**0.938**	**1.000**	**0.968**

(Y) updates the Memory Pool for future reuse, while a mismatch (N) triggers an escalation to the final verification tier.

Stage 4: Scholar Retrieval & Final Verification: The most resilient citations are handled by the Scholar Agent. It utilizes low-frequency, high-precision crawling to fetch the *Canonical Ground Truth*. The Judge Agent then performs a second-pass, character-level alignment. If this definitive check fails (N), the citation is flagged as Hallucinated with an associated provenance report; otherwise, it is successfully Verified and stored.

5 Experiments

5.1 Experimental Setup

In this section, we detail the implementation of our multi-agent framework and the hardware/software configurations used for the citation verification task.

Agent Implementation and Model Selection: All core reasoning and visual parsing tasks are powered by the state-of-the-art **Qwen3-VL-235B A22** model, deployed locally via the **vLLM** high-throughput serving framework to ensure data privacy and low-latency inference. The specific roles are instantiated as follows:

- **Planning Model (Orchestrator):** Implemented using Qwen3-VL-235B [9]. It acts as the central executor of the SOP, managing task states and thread-pool allocation. It first scans the initial and terminal 1,000 tokens of each PDF page to locate the "References" section precisely, then partitions the document for parallel processing.
- **Extractor Agent ($\mathcal{A}_{ext}$):** Utilizing the multimodal capabilities of Qwen3-VL-235B A22, this agent performs page-by-page OCR and structural parsing. It identifies individual citation strings and maps them into the predefined JSON metadata schema m_T, m_A, m_U, m_V.
- **Memory Agent ($\mathcal{A}_{mem}$):** Developed based on the **Mem0** [11] framework. It maintains a persistent, evolving knowledge graph of previously audited citations, enabling long-term context retention and rapid "fast-path" verification.
- **Web Search Agent ($\mathcal{A}_{web}$):** Integrated with the Google Search API for real-time evidence retrieval. It utilizes a deep-crawling protocol to bypass superficial snippets, ingesting full-text content from the top 5 search results for high-fidelity cross-referencing.
- **Judge Agent ($\mathcal{A}_{jud}$):** Powered by Qwen3-VL-235B, this agent executes the Strict Consistency Criterion (S_c). It performs character-level alignment between extracted metadata and external evidence, acting as the final arbiter for both preliminary and escalated verification stages.
- **Scholar Agent ($\mathcal{A}_{sch}$):** A specialized high-precision crawler [18] designed to interface with authoritative scholarly databases (e.g., Google Scholar) to fetch canonical ground-truth records ($\hat{M}_i$).

The detailed system prompts and standardized operating procedure (SOP) definitions for all agents are documented in Appendix D.

Infrastructure and Hyperparameters: The entire pipeline is executed on a high-performance compute cluster equipped with NVIDIA B200 GPUs. We employ a **Multi-thread Pool (Size=4)** to facilitate simultaneous citation auditing across multiple documents. For the visual-textual extraction, the model is configured with a temperature of 0.0 to ensure deterministic, exact-match parsing. The vector database for $\mathcal{A}_{mem}$ utilizes Cosine Similarity with a threshold of 0.92 to identify high-affinity citation matches.

Evaluation Metrics. To evaluate the performance of our citation verification system, we report both classification-based and efficiency-oriented metrics.

We define four types of prediction outcomes. **True Positives** refer to hallucinated citations that are correctly flagged as fake. **False Negatives** are hallucinated citations that are mistakenly predicted as real. **False Positives** denote real citations that are incorrectly flagged as fake, and **True Negatives** are real citations that are correctly identified as real.

Based on these outcomes, we compute standard metrics. **Accuracy** reflects the overall correctness of the predictions across all citations. **Precision** measures how often the citations flagged as hallucinations are actually hallucinated, indicating the system's ability to avoid false alarms. **Recall** quantifies how many of the actual hallucinated citations are successfully identified, capturing detection completeness. The **F1 score** balances precision and recall, offering a single measure of effectiveness in hallucination detection.

Table 4: Results on the real-world test set. Reveals a high level of statistical consistency with the generated test set

Model	Time	Price ($/1M tok)		Confusion Matrix				Metrics			
	/10 refs	In	Out	TP	FN	FP	TN	Acc	Prec	Rec	F1
Mixtral-8x7B-Instruct	2.3	0.60	0.60	95	372	757	2132	0.664	0.112	0.203	0.144
Llama-3.3-70B-Instruct	4.8	0.88	0.88	83	384	306	2583	0.794	0.213	0.178	0.194
Qwen3-Next-80B-A3B	3.7	0.15	1.50	231	233	1104	1785	0.601	0.173	0.498	0.257
Gemini-3-Pro	38.1	0.50	3.00	351	116	412	2477	0.843	0.460	0.752	0.571
GPT-5.2	48.8	1.75	14.00	366	101	1380	1510	0.559	0.210	0.784	0.331
GPTZero	26.2	70.00	0.00	338	129	1358	1531	0.557	0.199	0.724	0.312
Claude-Sonnet-4.5	13.3	3.00	15.00	349	118	756	2133	0.740	0.316	0.747	0.444
Our Model	**2.3**	**0.00**	**0.00**	**467**	**0**	**100**	**2789**	**0.972**	**0.823**	**1.000**	**0.903**

In addition to classification performance, we also assess system efficiency by measuring the average **runtime** required to verify a batch of 10 citations, along with the associated input/output costs. These metrics reflect the practical feasibility of large-scale deployment and help benchmark the trade-off between verification accuracy and computational cost.

5.2 Evaluation on Generated Benchmark

We evaluate all citation verification models on our generated benchmark, which consists of 3,586 real-world references and 2,500 hallucinated references produced through controlled perturbations spanning multiple error categories. In addition, we conduct evaluation on a real-world test set comprising 2,743 authentic references and 467 naturally occurring hallucinated citations collected from real scholarly sources.

Table 3 presents the performance of different citation verification models on our generated benchmark. Most existing models exhibit a clear unbalance between detecting hallucinated citations and preserving real references. For example, GPT 5.2 and Claude Sonnet 4.5 achieve strong recognition of real citations but still accept a non-negligible number of fabricated references as valid. Conversely, some open-source models detect more fake citations but introduce substantial false alarms on real references. In contrast, our model demonstrates strict and reliable detection of hallucinated citations, correctly identifying all fabricated references without false acceptance. At the same time, it maintains a low false-positive rate on real citations, preserving the majority of genuine references. This behavior indicates that our approach effectively enforces citation authenticity constraints rather than relying on permissive plausibility judgments, leading to substantially more trustworthy citation verification performance.

Table 3 also shows the runtime efficiency and API pricing comparison across representative citation-verification models. Notably, our framework achieves near-minimal processing time while incurring zero monetary cost, substantially outperforming commercial LLM-based solutions in cost efficiency. This advantage stems from our architectural design: large language models are used only for high-level planning and final judgment, rather than for deep thinking or retrieval. The majority of verification workload is handled by lightweight agents and external tools. As a result, our system maintains competitive latency while avoiding the substantial per-token pricing observed in proprietary models.

5.3 Evaluation on Real World Benchmark

We additionally assess citation verification performance on a real-world collection composed of authentic references and naturally occurring hallucinated citations drawn from scholarly manuscripts. Compared with the controlled benchmark setting, this evaluation reflects the ambiguity, noise, and incomplete metadata commonly encountered in practical academic writing scenarios.

The results in Table 4 reveal that existing approaches remain limited in balancing hallucination filtering with preservation of legitimate references. Methods that aggressively flag suspicious entries tend to over-reject genuine citations, whereas more permissive systems allow fabricated references to pass verification, indicating unresolved reliability challenges in real deployment conditions. In contrast, our framework achieves the strongest overall performance across all evaluation metrics. Specifically, it delivers the highest accuracy, precision, recall, and F1 score, with the F1 score exceeding that of the second-best system by more than 0.35 absolute points, indicating a substantial improvement in balanced verification capability. This margin demonstrates that our method detects hallucinated citations more reliably and maintains faithful acceptance of genuine references as well, overcoming the reliability limitations observed in prior systems.

Moreover, the consistency of relative model behavior between this real-world evaluation and the controlled benchmark suggests that the perturbation-based construction pipeline successfully captures the essential statistical characteristics of citation errors occurring in practice, further supporting the empirical validity and diagnostic value of our benchmark design.

5.4 Ablation Study

To rigorously assess the contribution of each module within our multi-agent architecture, we conduct an ablation study on the Real-World Benchmark. As detailed in Table 5, we evaluate three variants by selectively disabling key components: the Scholar Agent ($\mathcal{A}_{sch}$),

Table 5: **Ablation study on the Real-World Benchmark. We analyze the contribution of each component by removing them individually. "w/o Scholar Agent" denotes removing the final authoritative verification; "w/o Judge Agent" replaces the LLM judge with strict string matching; "w/o Web Search" relies solely on the Scholar Agent without preliminary web retrieval.**

Configuration	Time (s/10 refs)	Acc	Prec	Rec	F1
Full Framework	**2.3**	**0.970**	0.861	**1.000**	**0.925**
w/o Scholar Agent	1.9	0.915	**0.885**	0.684	0.772
w/o Judge Agent (Code)	0.8	0.604	0.225	1.000	0.367
w/o Web Search	18.4	0.942	0.810	0.955	0.877

the LLM-based Judge Agent ($\mathcal{A}_{jud}$), and the Web Search Agent ($\mathcal{A}_{web}$).

Impact of Authoritative Verification ($\mathcal{A}_{sch}$). Removing the Scholar Agent ("w/o Scholar Agent") leads to a precipitous drop in Recall from 1.000 to 0.684. This degradation indicates that relying solely on general web search leaves the system vulnerable to "resilient hallucinations"—fabricated references that appear plausible in general queries but are absent from canonical records. The Scholar Agent effectively acts as the final safety net, ensuring that only citations backed by authoritative bibliographic databases are accepted.

Necessity of Semantic Reasoning ($\mathcal{A}_{jud}$). Replacing the LLM-based Judge with a strict string-matching algorithm ("w/o Judge Agent") results in a catastrophic decline in Precision (0.861 → 0.225) and F1-score (0.925 → 0.367), despite maintaining perfect Recall. This reveals that rigid code-based verification is intolerant to real-world noise, such as abbreviation inconsistencies, minor typos, or formatting variations. The LLM Judge provides crucial *semantic resilience*, distinguishing between genuine metadata noise and actual hallucinations, thereby preventing a high rate of false alarms.

Efficiency of Cascade Retrieval ($\mathcal{A}_{web}$). Excluding the preliminary Web Search module ("w/o Web Search") causes the inference latency to skyrocket by approximately 8× (2.3s → 18.4s). Without the high-throughput filtering provided by $\mathcal{A}_{web}$, every query is forced to undergo the rate-limited, high-latency crawling process of the Scholar Agent. This confirms that the Web Search Agent serves as an essential "fast-path" filter, balancing system efficiency with verification rigor.

5.5 Additional Experiment Analysis

During experimentation, the suboptimal performance of advanced proprietary models appeared counterintuitive, motivating diagnostic evaluations via web-based LLM interfaces with observable reasoning behavior. We found that even when explicitly instructed to perform external retrieval, the systems do not reliably execute verifiable search procedures, and the provenance of implicitly retrieved evidence remains opaque. This black-box behavior makes retrieval neither enforceable nor transparently grounded, which is problematic for citation verification requiring explicit evidence tracing. This observation further underscores the necessity of specialized, auditable citation-verification tools that ground decisions in traceable external evidence, reinforcing the value of our benchmark and system as a principled alternative to reliance on closed-source general-purpose LLMs.

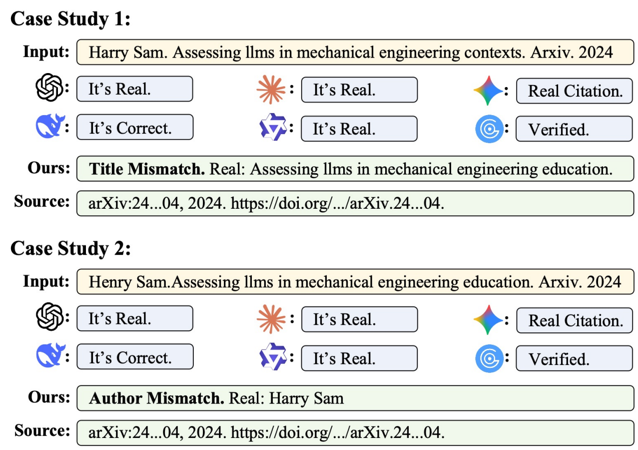

Figure 4: Case studies of citation verification

5.6 Case Study

We present representative case studies to qualitatively demonstrate the effectiveness and robustness of our citation verification framework. As illustrated in Figure 4, the system is capable of accurately identifying whether a citation refers to a real scholarly work, while further diagnosing fine-grained inconsistency types when mismatches occur.

In **Case Study 1**, although the queried citation corresponds to a real arXiv paper and is correctly recognized as non-hallucinated by multiple baseline systems, the cited title exhibits a subtle semantic deviation from the ground-truth record. Our framework successfully detects this *title mismatch*, retrieves the correct reference from arXiv, and explicitly reports the discrepancy, demonstrating sensitivity to partial but meaningful citation errors beyond binary real/fake classification. Similarly, in **Case Study 2**, the input citation again refers to an existing paper; however, the listed author name does not match the true authorship information. While existing tools label the citation as valid due to its overall plausibility, our system precisely identifies the *author mismatch* and recovers the correct author metadata from the authoritative source.

Across both examples, the framework not only distinguishes real citations from hallucinated ones, but also performs structured verification of individual metadata fields, including title and author information. By retrieving the correct ground-truth source and reporting explicit mismatch categories, our approach enables reliable, interpretable, and practically useful citation authenticity verification, which is essential for real-world scholarly and clinical documentation workflows.

6 Conclusion

As large language models become deeply integrated into scientific writing and peer-review workflows, the risk of hallucinated citations—references that are plausible yet nonexistent—poses a growing threat to research integrity. In this work, we take a decisive step toward addressing this challenge by introducing the first open, standardized, and scalable benchmark for hallucinated citation detection. Alongside this benchmark, we propose a novel multi-agent framework that operationalizes citation verification as a structured SOP-driven process, combining planning, retrieval, and judgment through specialized agents. Our framework demonstrates state-of-the-art performance in both controlled and real-world settings, outperforming leading commercial and open-source baselines across accuracy, efficiency, and interpretability. We envision our benchmark and system serving as foundational infrastructure for researchers, reviewers, and publishers to restore trust in citations and ensure scholarly accountability in the LLM era.

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A Citation Hallucination Taxonomy

To enable systematic and reproducible evaluation of citation verification systems, we propose a structured taxonomy that categorizes citation hallucinations according to the type and severity of bibliographic inconsistencies. Our taxonomy is grounded in naturally occurring citation errors observed in real scholarly manuscripts and is designed to capture both subtle and severe hallucination patterns generated by large language models.

A.1 Definition of Citation Hallucination

We define a *citation hallucination* as a bibliographic reference that appears plausible in form and content but fails to correspond to a valid scholarly record. Formally, given a citation metadata tuple $M = \{m_T, m_A, m_V, m_Y, m_{ID}\}$ representing title, authors, venue, year, and persistent identifiers (e.g., DOI or arXiv ID), a citation is considered *hallucinated* if no authoritative scholarly source can be found that simultaneously satisfies all essential metadata fields.

Unlike minor formatting inconsistencies or incomplete references, citation hallucinations involve semantic or factual mismatches that break the evidence chain between a claim and its cited source, thereby undermining scholarly attribution and verification.

A.2 Title-Level Hallucinations

Title-level hallucinations occur when the cited paper's title is incorrect or fabricated, while other metadata fields remain partially or fully plausible. This error type is particularly challenging because minor lexical variations or paraphrases may appear legitimate to both humans and automated systems.

We identify three representative subclasses:

- **Keyword Substitution:** Core technical terms in the original title are replaced with semantically related but incorrect alternatives.
- **Paraphrased Fabrication:** The title is fluently rephrased into a plausible variant that does not correspond to any existing publication.
- **Topic-Conditioned Synthesis:** A realistic-looking title is generated within the same research area without grounding in any real work.

A citation exhibiting any of the above patterns is labeled as hallucinated if no exact or canonical title match can be verified in authoritative bibliographic databases.

A.3 Author-Level Hallucinations

Author-level hallucinations involve incorrect or fabricated authorship information. These errors directly distort scholarly attribution and can mislead citation-based evaluation and credit assignment.

We categorize author hallucinations into four types:

- **Author Addition:** Nonexistent or unrelated author names are inserted into an otherwise valid author list.
- **Author Deletion:** One or more legitimate authors are omitted from the citation.
- **Name Perturbation:** Author names are altered through spelling changes, reordering of given and family names, or partial truncation.
- **Fully Fabricated Authorship:** The entire author list does not correspond to any real publication.

A citation is marked as hallucinated if the author list cannot be aligned with a verified scholarly record under standard name normalization rules.

A.4 Metadata-Level Hallucinations

Metadata-level hallucinations refer to inconsistencies in bibliographic fields beyond title and authors, including venue, publication year, and persistent identifiers.

We consider the following categories:

- **Venue Mismatch:** The cited venue does not match the actual publication outlet of the work.
- **Year Mismatch:** The publication year is incorrect beyond acceptable preprint-to-final-version variation.
- **Identifier Fabrication:** DOI, arXiv ID, or other persistent identifiers are invalid or mismatched.

Such hallucinations are particularly difficult to detect through surface-level similarity matching, as the citation may appear structurally complete despite being bibliographically invalid.

A.5 Compound and Cross-Field Hallucinations

In practice, citation hallucinations often involve multiple simultaneous inconsistencies. We therefore include a compound category for citations that exhibit errors across two or more metadata fields (e.g., both title and authors).

These compound hallucinations represent the most severe form of citation fabrication and are always labeled as hallucinated in our benchmark.

A.6 Relation to Real-World Citation Errors

Our taxonomy is informed by empirical analysis of citation errors observed in real conference and journal submissions. We emphasize that not all citation inaccuracies constitute hallucinations. Minor formatting variations, missing page numbers, or capitalization differences are not considered hallucinations as long as the core bibliographic identity remains verifiable.

This distinction ensures that the benchmark targets genuinely harmful hallucinations rather than penalizing benign citation noise, thereby supporting fair and realistic evaluation of citation verification systems.

B Dataset Construction and Statistics

This appendix details the construction process and statistical composition of the generated citation hallucination dataset used in our benchmark. The dataset is designed to provide controlled, fine-grained evaluation of citation verification systems under diverse and realistic hallucination scenarios.

B.1 Source of Real Citations

We begin by collecting a pool of verified real citations from publicly available bibliographic repositories and open-access scholarly sources. Each citation is extracted in structured BibTeX format and manually verified to ensure correctness across all essential metadata fields, including title, authors, venue, publication year, and persistent identifiers (e.g., DOI or arXiv ID). These verified citations serve as the ground-truth references from which hallucinated variants are systematically generated.

B.2 Hallucinated Citation Generation

Building on the verified citation pool, we generate hallucinated references through controlled perturbations guided by the taxonomy described in Appendix A. Each hallucinated citation is derived from a real reference by modifying one or more metadata fields while preserving overall plausibility and structural validity.

Specifically, the generated test set consists of three primary hallucination categories:

- **Title Errors (1000 instances):** The original title is replaced with a semantically plausible but invalid variant, while all other metadata fields remain unchanged. These errors are constructed via keyword substitution, paraphrasing-based rewriting, and topic-conditioned synthesis.
- **Author Errors (1000 instances):** The author list is perturbed through controlled operations, including author addition, deletion, name-level perturbation, or full authorship fabrication, while the remaining metadata fields are preserved.
- **Metadata Errors (500 instances):** Non-title and non-author bibliographic fields are modified, including venue mismatches, publication year shifts, and fabricated or mismatched persistent identifiers.

Each hallucinated citation maintains valid BibTeX structure and formatting to prevent trivial detection based on syntactic irregularities.

B.3 Dataset Composition and Balance

The final generated test set contains a total of 2,500 hallucinated citations, distributed across error categories as summarized above. To enable balanced evaluation, these hallucinated citations are paired with an equal number of verified real citations sampled from the original reference pool. This results in a generated benchmark comprising 5,000 citation instances in total, evenly split between real and hallucinated references.

The controlled distribution across hallucination types ensures that evaluation results are not dominated by a single error mode and allows for targeted analysis of model robustness to different classes of citation hallucinations.

B.4 Quality Control and Validation

All generated hallucinated citations are subjected to automated retrieval-based checks followed by manual validation by the author team. A hallucinated citation is retained in the benchmark only if no authoritative scholarly record can be found that matches the perturbed metadata under standard normalization rules. This verification process ensures that the generated dataset faithfully simulates realistic citation hallucinations observed in real-world academic writing, while avoiding false positives arising from incomplete indexing or transient database inconsistencies.

C Human Annotation and Verification Protocol

To ensure the reliability and correctness of the benchmark labels, all citation annotations in both the generated and real-world datasets were conducted exclusively by the author team. No crowd-sourced annotators or automated labeling pipelines were used at any stage of the annotation process.

C.1 Annotation Scope

Each citation instance is represented as a structured metadata tuple, including title, authors, venue, publication year, and persistent identifiers when available. Annotators are tasked with determining whether the citation corresponds to a valid scholarly record and whether all critical metadata fields are consistent with authoritative sources.

C.2 Verification Procedure

For each citation, annotators perform a manual verification process consisting of the following steps:

- **Authoritative Search:** The citation is queried against authoritative scholarly databases, including Google Scholar, publisher websites, and institutional repositories.
- **Metadata Cross-Checking:** Retrieved records are manually compared against the citation's title, author list, venue, year, and identifiers to assess consistency.
- **Existence Validation:** A citation is marked as *Real* only if a corresponding scholarly work can be confidently identified with matching core metadata.
- **Hallucination Confirmation:** A citation is labeled as *Hallucinated* if no matching scholarly record can be found or if one or more critical metadata fields are demonstrably inconsistent.

C.3 Labeling Criteria

We adopt a strict but realistic labeling standard. Minor formatting variations, capitalization differences, or missing optional fields (e.g., page numbers) are not considered hallucinations as long as the core bibliographic identity remains verifiable. Conversely, any inconsistency in essential fields—such as a non-existent title, incorrect authorship, invalid venue attribution, or fabricated identifiers—results in a hallucinated label.

C.4 Conflict Resolution

All citation instances are independently reviewed by at least two authors. In cases of disagreement or ambiguity, the citation is jointly re-examined by the author team until a consensus decision is reached. Citations for which no consensus can be achieved after exhaustive verification are excluded from the benchmark to avoid introducing label noise.

C.5 Quality Assurance

To further ensure annotation integrity, a subset of citations is randomly re-checked after the initial labeling process to verify consistency across annotators. This manual auditing step helps prevent systematic bias and ensures uniform application of the labeling criteria throughout the dataset.

D Prompt Templates and SOP Details

This appendix documents the core prompt templates used in our SOP-driven multi-agent citation verification framework. To ensure reproducibility and transparency, we explicitly disclose the prompts for the Planning (Orchestrator) Agent and the Judge Agent, which together govern task execution and final verification decisions.

D.1 Planning (Orchestrator) Prompt

The Planning Agent serves as the central SOP executor. Its responsibility is not to verify citations directly, but to determine the execution path for each citation according to the predefined verification stages (memory lookup, web verification, scholar verification). The agent outputs a structured plan specifying which agents to invoke and in what order.

System Prompt (Planning Agent).

> You are a citation verification orchestrator. Your task is to execute a strict Standardized Operating Procedure (SOP) for verifying academic citations.
> You do NOT judge citation correctness yourself. You ONLY decide which verification step to run next for each citation.
> Available verification stages: 1. Memory Lookup (fast-path): check whether this citation has been previously verified. 2. Web Verification: compare the citation against downloaded content from web search. 3. Scholar Verification: perform authoritative verification using scholarly databases.
> SOP RULES: - Always attempt Memory Lookup first. - If Memory Lookup confirms the citation as verified, STOP. - If Memory Lookup fails or returns unknown, proceed to Web Verification. - If Web Verification confirms a valid match, STOP and mark as verified. - If Web Verification fails or is inconclusive, escalate to Scholar Verification. - Scholar Verification is the FINAL decision stage.
> Constraints: - Follow the SOP strictly. - Do not skip stages. - Do not perform reasoning beyond task routing. - Do not modify citation metadata.

Planning Output Format. The Planning Agent outputs a JSON plan for each citation:

```
{
  "citation_id": <id>,
  "next_action": "memory" | "web" | "scholar" | "stop",
  "reason": "brief justification"
}
```

This structured output enables deterministic execution and multi-threaded scheduling in the verification pipeline.

D.2 Judge Agent Prompt

The Judge Agent performs strict citation-to-evidence matching. It receives the citation metadata and the downloaded content from the top search results, and determines whether a valid match exists according to explicitly defined rules.

Judge Prompt.

Compare the citation with the downloaded content from search results and determine if any matches.
Citation to verify: Title: {title} Authors: {authors} Venue: {venue} Year: {year}
Downloaded content from top 5 search results: {documents}
Return JSON:

```
Return JSON:
{
    "match": true/false,
    "matched_result": <number or null>,
    "note": "brief reason"
}
```

MATCHING CRITERIA: 1. Title must match COMPLETELY. - All significant words must be present. - Ignore only case, punctuation, and articles ("a", "an", "the").
2. Authors must match COMPLETELY. - ALL authors listed in the citation must appear in the downloaded content. - Author order may differ, but all names must be present.
3. Venue matching rules: - If citation venue is "arXiv" OR downloaded content is from arXiv → ALLOW any venue difference. - If citation venue is a conference (e.g., NeurIPS, ICML, CVPR) AND downloaded content is from a DIFFERENT conference → REJECT. - If citation venue is conference and downloaded content is journal (or vice versa) → ALLOW. - Year differences are acceptable if the work is the same (e.g., arXiv 2020, conference 2021).
IMPORTANT: - match = true if title AND authors match AND venue rules are satisfied. - match = false if ANY of the following holds: - Title does not match. - Authors do not fully match. - Venue rule is violated. - Do NOT guess. - Do NOT use partial matches. - Be strict and conservative.

D.3 Role Separation and Determinism

The Planning Agent and Judge Agent have strictly separated responsibilities. The Planning Agent performs task routing only and never evaluates citation correctness. The Judge Agent performs deterministic matching under explicit rules and never controls pipeline execution. This separation ensures that citation verification decisions are auditable, reproducible, and not conflated with heuristic reasoning or uncontrolled generation.